Assignment 3

Data Management (INFO125)

Deadline: Friday, 18.10.2024 at 13:00

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Question 1

Consider the following relations for a database that keeps track of booking flats and choose the primary keys:

FLAT (Flat#, Type, Address, Price_perUnit)

OPTION (Flat#, Option_name, Extra_price)

BOOKING (Agent#, Flat#, Date, Booking_price)

AGENT (Agent_id, Name, Phone)

FLAT

Flat#	Type	Address	Price_perUnit
A105	3BHK	Waltham Abbey	200
A308	5BHK	Charlotte Street	220
B216	3BHK	Old Gloucester Street	290
B201	3BHK	Old Gloucester Street	290

OPTION

Flat#	Option_name	Extra_price
A105	Balcony East Facing	1100
A308	Window	600
B216	Mezzanine Floor	1000

BOOKING

Agent#	Flat#	Date	Booking_price
A1	A105	12-01-2016	1100
A2	B216	02-02-2016	2100

AGENT

Agent_id	Name	Phone
A1	Will Smith	44 20 7520 1490
A2	Ashley Rawdon	44 20 8650 2999
A3	John Stuart	44 20 7419 5000

Now answer the following questions:

a) What can be the foreign keys for this schema, stating any assumptions you make.

   a Flat# in OPTION references Flat# in FLAT. The OPTION table lists options for flats, and each option is associated with a spesific flat.

   b Agent# in BOOKING references Agent_id in AGENT. Each booking is made by an agent.

   c Flat# in BOOKING references Flat# in FLAT. Each booking is for a spesific flat. Flat# in BOOKING should match a flat listed in the FLAT table.

b) Which of the following operations violate the referential integrity constraints and which one does not.

- INSERT <'A1', 'A308', '03-03-2016', 2100> INTO BOOKING
    - Agent# = 'A1' exists in the AGENT table.
    - Flat# = 'A308' exists in the FLAT table.
    - No violation.
- INSERT <'A1', 'C205', '03-03-2016', 2100> INTO BOOKING
    - Agent# = 'A1' exists in the AGENT table.
    - Flat# = 'C205' does not exist in the Flat table.

- This is a violation because the required flat doesn´t exist in the Flat table.

- INSERT <’A4’, ‘Arilia Sengupta’, ‘44 20 7836 5454’> INTO AGENT
    - No violation because it doesn´t reference any other table.

Question 2

Consider the following Hotel, Room, Booking and Guest schemas in a DBMS.

- Hotel(hotelNo,hotelName, hotelType, hotelAddress, hotelCity, numRoom)
- Room(roomNo, hotelNo, roomPrice)
- Booking(bookingNo, hotelNo, guestNo, checkIn, checkOut, totalGuest, roomNo)
- Guest(guestNo, firstName, lastName, guestAddress)

The hotelNo is the primary key for Hotel table and roomNo is the primary key for the Room relation. Booking stores the details of room reservations and bookingNo is the primary key. Guest stores the guests details and guestNo is the primary key.

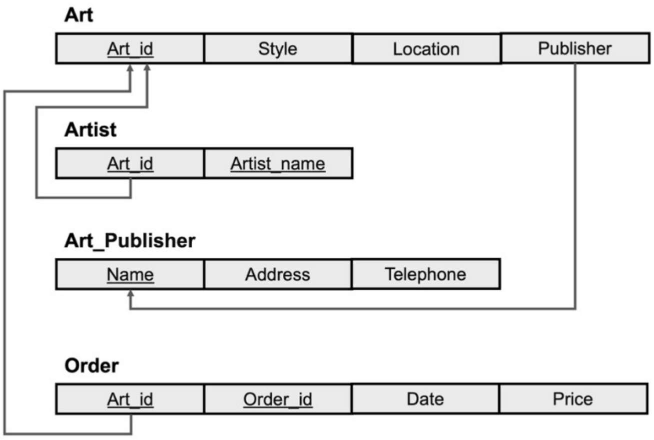
a) Write SQL script to list details of hotels in New York in descending order by number of rooms

b) Write SQL script to list the details of all guests whose first name is ‘Sara’ in ascending order by address.

```sql
SELECT hotelNo, hotelName, hotelType, hotelAddress, hotelCity, numRoom
FROM Hotel
WHERE hotelCity = 'New York'
ORDER BY numRoom DESC;

SELECT guestNo, firstName, lastName, guestAddress
FROM Guest
WHERE firstName = 'Sara'
ORDER BY guestAddress ASC;


```

Question 3

The following figure represents a database schema called Art_Gallery. Write SQL statements that would create a Art_Gallery database with the given tables (relations) and columns (attributes). Specify the primary keys and foreign keys according to the figure. You can make your own decisions on the data types for the columns and their domains. You can also decide whether or not certain columns can have NULL values. Note: Different Solutions are possible.

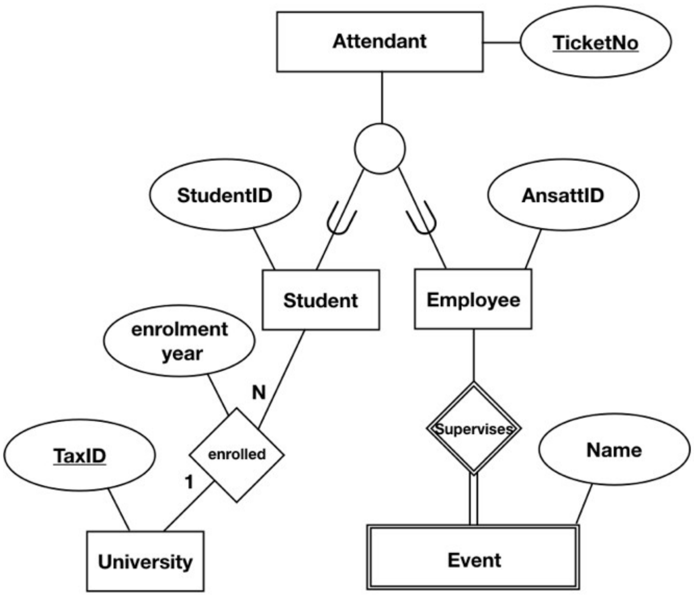


Art_Gallery

```sql
CREATE DATABASE Art_Gallery;
USE Art_Gallery;

CREATE TABLE Art_Publisher (
    Name VARCHAR(100) PRIMARY KEY,
    Address VARCHAR(100) NOT NULL,
    Telephone VARCHAR(8) NOT NULL
);

CREATE TABLE Art (
    Art_id INT PRIMARY KEY AUTO_INCREMENT,
    Style VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL,
    Publisher VARCHAR(100),
    FOREIGN KEY (Publisher) REFERENCES Art_Publisher(Name)
        ON DELETE SET NULL
);

CREATE TABLE Artist (
    Art_id INT,
    Artist_name VARCHAR(100) NOT NULL,
    PRIMARY KEY (Art_id),
    FOREIGN KEY (Art_id) REFERENCES Art(Art_id)
        ON DELETE CASCADE
);

CREATE TABLE `Order` (
    Order_id INT PRIMARY KEY AUTO_INCREMENT,
    Art_id INT,
    Date DATE NOT NULL,
    Price DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (Art_id) REFERENCES Art(Art_id)
        ON DELETE CASCADE
);
```

Question 4

Consider the following EER diagram and map it to the relational model as well as indicating the foreign keys.



Note: Different Solutions are possible